

RAVI RAJ

Backend Developer

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Summary

Backend developer with hands-on experience in relational schema design, REST API development, and query optimization across Python systems. Builds data-driven applications end-to-end, from constraint-enforced database architecture to CRUD systems, applying optimization techniques to reduce processing overhead, including a fault-repair framework that cut model recovery time without retraining.

Technical Skills

- **Languages & Core CS:** Python, C++ · Data Structures & Algorithms · Object-Oriented Design · Operating Systems
- **Backend & APIs:** REST API Design (Flask, FastAPI) · Schema Design · Query Optimization · CRUD Architecture
- **Databases:** MySQL, SQL Server, SQLite – relational modeling, foreign-key constraint design, indexing
- **Engineering Practice:** Git/GitHub version control · Agile/SDLC · Unit Testing · Debugging under production-like constraints

Key Projects

Inventory Management & Database Pipeline [\[GitHub\]](#)

Python | *SQL* | *SQLite* | *MySQL* | *Git* | *GitHub*

- Architected a 5-table relational schema with enforced foreign-key constraints, eliminating data integrity issues across inventory records.
- Built Python-based CRUD operations and optimized SQL queries to support real-time stock tracking.
- Added automated low-stock alerting and modular CLI tooling, version-controlled via Git/GitHub.

Autonomous DNN Repair Using Gradient Detection and LoRA [\[GitHub\]](#)

Python | *NumPy* | *SciPy*

- Engineered an automated fault-detection and repair framework for neural networks using gradient-based diagnostics and LoRA fine-tuning — avoiding full retraining while restoring model reliability.
- Achieved a 15× improvement in repair speed through targeted algorithm optimization.
- Applied systematic debugging and performance testing to improve overall model reliability and maintainability.

Handwritten Devanagari Character Recognition Using SCNN [\[GitHub\]](#)

Python | *OpenCV* | *Flask* | *SQL Server*

- Built an SCNN-based recognition system achieving 93.91% accuracy across 92,000+ handwritten character images.
- Designed image preprocessing pipelines and Flask REST APIs to support real-time character prediction.
- Refined database queries and model logic through systematic testing, improving overall system reliability.

Education

Chandigarh University, Mohali 2022 – 2026

B.E., Computer Science & Business Systems CGPA: 7.94

Certifications

- Oracle Cloud Infrastructure Certified – Data Science
- Tata (Forage): Generative AI-Powered Data Analytics Job Simulation

Leadership & Involvement

- **CSBIZ Nexus Society (2024–2025)** – Coordinated technical workshops with industry professionals and supported society operations